Honda Digital Service Development - 2024 Summer Intern Project Report

Enhancing in-car interaction with On-Device Generative AI

Overview

This project demonstrates the capabilities of generative AI, utilizing the Llama 3-8B model on the NVIDIA Jetson Orin Nano 8GB, which serves as the benchmark for hardware of the latest in-car head unit chips. The AI was configured to simulate interactions as a Honda vehicle, with future possibility to integrate real-time vehicle data for enhanced interactivity. The model can be optimized or replaced with alternatives to reduce RAM usage, ensuring a lightweight solution. This project highlights the significant potential of generative AI in automotive technology, serving as an essential benchmark as the industry moves forward.

Consider this question:

In what ways could **Generative AI enhance** the overall user experience in **vehicle interactions?**

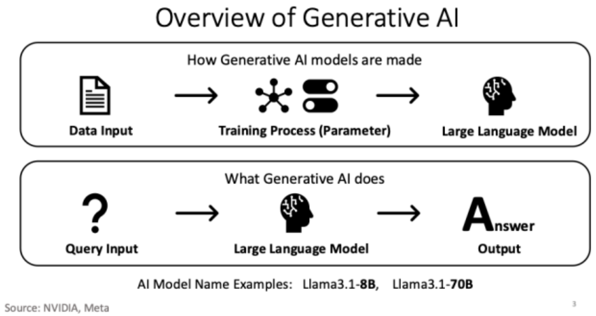
Ryo Fujimura

Jake Kwon

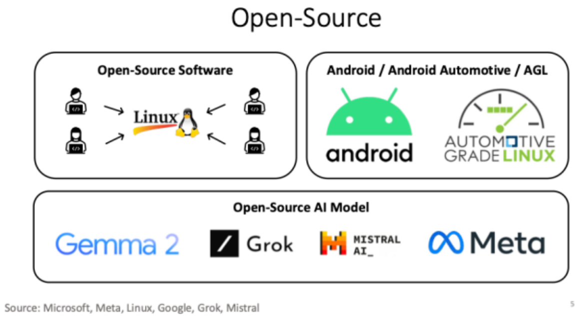
Key Project Factors

Project Owner: Ryo Fujimura
Supervisor: Jake Kwon
Duration: 8 weeks (Summer 2024)
Hardware: NVIDIA Jetson Orin Nano 8GB
Software: Linux, JetPack, Llama3-8B

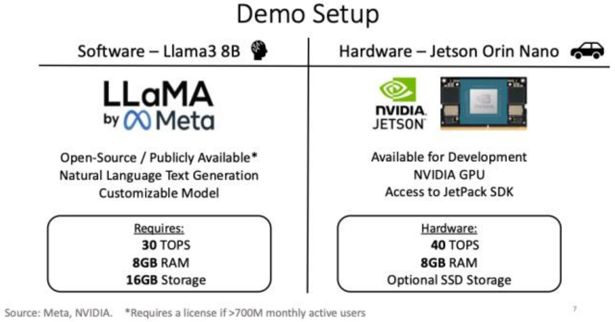
Presentation Highlights



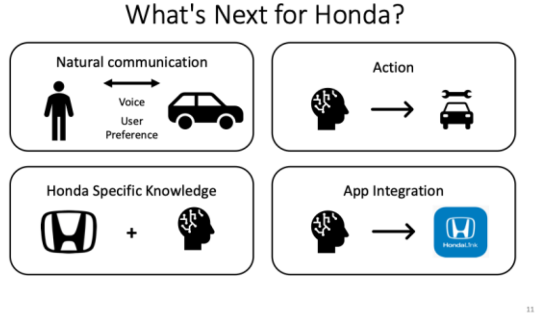
The presentation starts with an overview of generative AI, ensuring the audience had a shared understanding of the concept.



Open-source platforms like Android and AGL drive automotive innovation. Honda must adopt AI open-source to remain competitive.



Deploying the Llama 3 8B model on Jetson Orin Nano with 8GB RAM, serving as a benchmark for the latest Qualcomm Automotive Chip.



Honda's next step with generative AI includes natural voice, Honda-specific knowledge, action calling, and app integration and more.